COMP4670/COMP8600
Statistical Machine Learning - Sem 1 2022

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1 INTRODUCTION

polynomial curve fitting

$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

error function: mean squared error (MSE)

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \{y(x_n, \mathbf{w}) - t_n\}^2$$

root-mean-square (RMS) error

$$E_{\mathrm{RMS}} = \sqrt{2E(\mathbf{w}^\star)/N}$$

Tikhonov regularization; shrinkage (statistics); ridge regression; weight decay (neural network)

$$\widetilde{E}(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \{y(x_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

the squared Euclidean norm of the coefficients

$$\|\mathbf{w}\|^2 \equiv \mathbf{w}^{\mathrm{T}} \mathbf{w} = w_0^2 + w_1^2 + \ldots + w_M^2$$

sum rule of probability

$$p(X) = \sum_{Y} p(X, Y) \text{ or } p(x) = \int p(x, y) \mathrm{d}y$$

product rule of probability

$$p(X, Y) = p(Y \mid X) p(X) \text{ or } p(x, y) = p(y \mid x) p(x)$$

conditional probability

$$p(Y \mid X) = \frac{p(X \mid Y) p(Y)}{p(X)}$$

Kolmogorov axioms; probability axioms

$$p(x) \geqslant 0$$
$$\int_{-\infty}^{\infty} p(x) \mathrm{d}x = 1$$

Jacobian transform; Jacobian factor

$$\begin{aligned} x &= g(y) \\ p_y(y) &= p_x(x) \left| \frac{\mathrm{d}x}{\mathrm{d}y} \right| \\ &= p_x(g(y)) \left| g'(y) \right| \end{aligned}$$

$$\begin{aligned} &\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \mathbf{y} \sim \mathbf{A}\mathbf{x} + \mathbf{b} \\ &\Rightarrow \mathbf{y} \sim \mathcal{N}\left(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top\right) \end{aligned}$$

expectation of $f(x)$

$$\mathbb{E}[f] = \sum_x p(x) f(x) = \int p(x) f(x) \mathrm{d}x$$

variance of $f(x)$

$$\mathrm{var}[f] = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])^2\right]$$

covariance of x and y

$$\begin{aligned} \mathrm{cov}[x, y] &= \mathbb{E}_{x,y}[\{x - \mathbb{E}[x]\}\{y - \mathbb{E}[y]\}] \\ &= \mathbb{E}_{x,y}[xy] - \mathbb{E}[x]\mathbb{E}[y] \end{aligned}$$

$$\begin{aligned} \mathrm{cov}[\mathbf{x}, \mathbf{y}] &= \mathbb{E}_{\mathbf{x},\mathbf{y}}\left[\{\mathbf{x} - \mathbb{E}[\mathbf{x}]\}\left\{\mathbf{y}^{\mathrm{T}} - \mathbb{E}\left[\mathbf{y}^{\mathrm{T}}\right]\right\}\right] \\ &= \mathbb{E}_{\mathbf{x},\mathbf{y}}\left[\mathbf{x}\mathbf{y}^{\mathrm{T}}\right] - \mathbb{E}[\mathbf{x}]\mathbb{E}\left[\mathbf{y}^{\mathrm{T}}\right] \end{aligned}$$

Bayes' theorem

$$p(\mathbf{w} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \mathbf{w}) p(\mathbf{w})}{p(\mathcal{D})}$$

$$\text{posterior} \ \propto \ \text{likelihood} \ \times \ \text{prior}$$

Gaussian distribution

$$\mathcal{N}\left(x \mid \mu, \sigma^2\right) = \frac{1}{\left(2\pi\sigma^2\right)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}$$

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} \mathcal{N}\left(x \mid \mu, \sigma^2\right) x \ \mathrm{d}x = \mu$$

$$\mathbb{E}\left[x^2\right] = \int_{-\infty}^{\infty} \mathcal{N}\left(x \mid \mu, \sigma^2\right) x^2 \ \mathrm{d}x = \mu^2 + \sigma^2$$

$$\mathrm{var}[x] = \mathbb{E}\left[x^2\right] - \mathbb{E}[x]^2 = \sigma^2$$

Gaussian integral

$$\int_{-\infty}^{\infty} e^{-a(x+b)^2} \, dx = \sqrt{\frac{\pi}{a}}$$

multivariate Gaussian distribution

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}} \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$$

independent and identically distributed Gaussian random variables

$$p\left(\mathbf{x} \mid \mu, \sigma^2\right) = \prod_{n=1}^{N} \mathcal{N}\left(x_n \mid \mu, \sigma^2\right)$$

the log likelihood of Gaussian random variables

$$\ln p\left(\mathbf{x} \mid \mu, \sigma^2\right) = -\frac{1}{2\sigma^2} \sum_{n=1}^{N} \left(x_n - \mu\right)^2 - \frac{N}{2} \ln \sigma^2 - \frac{N}{2} \ln(2\pi)$$

the maximum likelihood solution

$$\mu_{\mathrm{ML}} = \frac{1}{N} \sum_{n=1}^{N} x_n$$

$$\sigma^2_{\mathrm{ML}} = \frac{1}{N} \sum_{n=1}^{N} \left(x_n - \mu_{\mathrm{ML}}\right)^2$$

population mean and population variance

$$\begin{gathered} \mathbb{E}\left[\mu_{\mathrm{ML}}\right] = \mu \\ \mathbb{E}\left[\sigma^2_{\mathrm{ML}}\right] = \left(\tfrac{N-1}{N}\right) \sigma^2 \end{gathered}$$

unbiased estimate for the variance parameter

$$\widetilde{\sigma}^2 = \frac{N}{N-1} \sigma^2_{\mathrm{ML}} = \frac{1}{N-1} \sum_{n=1}^{N} \left(x_n - \mu_{\mathrm{ML}}\right)^2$$

polynomial curve fitting

$$p(t \mid x, \mathbf{w}, \beta) = \mathcal{N}\left(t \mid y(x, \mathbf{w}), \beta^{-1}\right)$$

the log likelihood function

$$\ln p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) = -\frac{\beta}{2} \sum_{n=1}^{N} \left\{y\left(x_n, \mathbf{w}\right) - t_n\right\}^2 + \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi)$$

the maximum likelihood solution of the precision parameter β

$$\frac{1}{\beta_{\mathrm{ML}}} = \frac{1}{N} \sum_{n=1}^{N} \left\{y\left(x_n, \mathbf{w}_{\mathrm{ML}}\right) - t_n\right\}^2$$

maximizing likelihood is equivalent, so far as determining $\mathbf{w}$ is concerned, to minimizing the sum-of-squares error function

$$\mathbf{w}_{\mathrm{ML}} \equiv \arg\min_{\mathbf{w}} E(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\left\{y\left(x_n, \mathbf{w}\right) - t_n\right\}^2$$

predictive distribution via the maximum likelihood parameters

$$p\left(t \mid x, \mathbf{w}_{\mathrm{ML}}, \beta_{\mathrm{ML}}\right) = \mathcal{N}\left(t \mid y\left(x, \mathbf{w}_{\mathrm{ML}}\right), \beta_{\mathrm{ML}}^{-1}\right)$$

prior distribution over the polynomial coefficients $\mathbf{w}$

$$p(\mathbf{w} \mid \alpha) = \mathcal{N}\left(\mathbf{w} \mid \mathbf{0}, \alpha^{-1}\mathbf{I}\right) = \left(\frac{\alpha}{2\pi}\right)^{(M+1)/2} \exp\left\{-\frac{\alpha}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w}\right\}$$

the posterior distribution for $\mathbf{w}$

$$p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) \propto p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) p(\mathbf{w} \mid \alpha)$$

maximum a posteriori probability (MAP) estimate for $\mathbf{w}$
maximizing the posterior distribution is equivalent to minimizing the regularized sum-of-squares error function, where $\lambda = \alpha/\beta$

$$\mathbf{w}_{\mathrm{MAP}} \equiv \arg\min_{\mathbf{w}} \frac{\beta}{2}\sum_{n=1}^{N}\left\{y\left(x_n, \mathbf{w}\right) - t_n\right\}^2 + \frac{\alpha}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w}$$

predictive distribution (Gaussian)

$$p(t \mid x, \mathbf{x}, \mathbf{t}) = \int p(t \mid x, \mathbf{w}) p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}) \mathrm{d}\mathbf{w}$$

$$p(t \mid x, \mathbf{x}, \mathbf{t}) = \mathcal{N}\left(t \mid m(x), s^2(x)\right)$$

$$m(x) = \beta \boldsymbol{\phi}(x)^{\mathrm{T}} \mathbf{S} \sum_{n=1}^{N} \boldsymbol{\phi}\left(x_n\right) t_n$$

$$s^2(x) = \beta^{-1} + \boldsymbol{\phi}(x)^{\mathrm{T}} \mathbf{S} \boldsymbol{\phi}(x)$$

$$\mathbf{S}^{-1} = \alpha \mathbf{I} + \beta \sum_{n=1}^{N} \boldsymbol{\phi}\left(x_n\right) \boldsymbol{\phi}(x)^{\mathrm{T}}$$

$$\boldsymbol{\phi}(x) := \phi_i(x) = x^i \text{ for } i = 0, \ldots, M$$

Akaike information criterion (AIC)

$$\ln p\left(\mathcal{D} \mid \mathbf{w}_{\mathrm{ML}}\right) - M$$

the expected loss

$$\mathbb{E}[L] = \iint L(t, y(\mathbf{x})) p(\mathbf{x}, t) \mathrm{d}\mathbf{x} \mathrm{d}t$$

$$\mathbb{E}[L] = \iint \{y(\mathbf{x}) - t\}^2 p(\mathbf{x}, t) \mathrm{d}\mathbf{x} \mathrm{d}t$$

Minkowski loss

$$\mathbb{E}\left[L_q\right] = \iint |y(\mathbf{x}) - t|^q p(\mathbf{x}, t) \mathrm{d}\mathbf{x} \mathrm{d}t$$

entropy

$$\mathrm{H}[p] = -\sum_i p\left(x_i\right) \ln p\left(x_i\right)$$

differential entropy

$$\mathrm{H}[x] = -\int p(x) \ln p(x) \mathrm{d}x \quad \mathrm{H}[\mathbf{x}] = -\int p(\mathbf{x}) \ln p(\mathbf{x}) \mathrm{d}\mathbf{x}$$

Stirling's approximation

$$\ln N! \simeq N \ln N - N$$

conditional entropy

$$\mathrm{H}[\mathbf{y} \mid \mathbf{x}] = -\iint p(\mathbf{y}, \mathbf{x}) \ln p(\mathbf{y} \mid \mathbf{x}) \mathrm{d}\mathbf{y} \mathrm{d}\mathbf{x}$$

$$\mathrm{H}[\mathbf{x}, \mathbf{y}] = \mathrm{H}[\mathbf{y} \mid \mathbf{x}] + \mathrm{H}[\mathbf{x}]$$

Kullback-Leibler divergence (relative entropy)

$$\begin{aligned} \mathrm{KL}(p \| q) &= -\int p(\mathbf{x}) \ln q(\mathbf{x}) \mathrm{d}\mathbf{x} - \left(-\int p(\mathbf{x}) \ln p(\mathbf{x}) \mathrm{d}\mathbf{x}\right) \\ &= -\int p(\mathbf{x}) \ln \left\{\frac{q(\mathbf{x})}{p(\mathbf{x})}\right\} \mathrm{d}\mathbf{x} \end{aligned}$$

$$\mathrm{KL}(p \| q) \not\equiv \mathrm{KL}(q \| p)$$

Jensen's inequality

$$f\left(\sum_{i=1}^M \lambda_i x_i\right) \leqslant \sum_{i=1}^M \lambda_i f\left(x_i\right)$$

$$f(\mathbb{E}[x]) \leqslant \mathbb{E}[f(x)]$$

$$f\left(\int \mathbf{x} p(\mathbf{x}) \mathrm{d}\mathbf{x}\right) \leqslant \int f(\mathbf{x}) p(\mathbf{x}) \mathrm{d}\mathbf{x}$$

$$\mathrm{KL}(p \| q) = -\int p(\mathbf{x}) \ln \left\{\frac{q(\mathbf{x})}{p(\mathbf{x})}\right\} \mathrm{d}\mathbf{x} \geqslant -\ln \int q(\mathbf{x}) \mathrm{d}\mathbf{x} = 0$$

mutual information

$$\begin{aligned} \mathrm{I}[\mathbf{x}, \mathbf{y}] &\equiv \mathrm{KL}(p(\mathbf{x}, \mathbf{y}) \| p(\mathbf{x}) p(\mathbf{y})) \\ &= -\iint p(\mathbf{x}, \mathbf{y}) \ln \left(\frac{p(\mathbf{x}) p(\mathbf{y})}{p(\mathbf{x}, \mathbf{y})}\right) \mathrm{d}\mathbf{x} \mathrm{d}\mathbf{y} \end{aligned}$$

$$\mathrm{I}[\mathbf{x}, \mathbf{y}] = \mathrm{H}[\mathbf{x}] - \mathrm{H}[\mathbf{x} \mid \mathbf{y}] = \mathrm{H}[\mathbf{y}] - \mathrm{H}[\mathbf{y} \mid \mathbf{x}]$$

2 PROBABILITY DISTRIBUTIONS

Bernoulli distribution

$$\mathrm{Bern}(x \mid \mu) = \mu^x (1-\mu)^{1-x}$$

binomial distribution

$$\mathrm{Bin}(m \mid N, \mu) = \binom{N}{m} \mu^m (1-\mu)^{N-m}$$

$$\mathbb{E}[m] \equiv \sum_{m=0}^{N} m \,\mathrm{Bin}(m \mid N, \mu) = N\mu$$

$$\mathrm{var}[m] \equiv \sum_{m=0}^{N} (m - \mathbb{E}[m])^2 \,\mathrm{Bin}(m \mid N, \mu) = N\mu(1-\mu)$$

beta distribution

$$\mathrm{Beta}(\mu \mid a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}$$

$$\mathbb{E}[\mu] = \frac{a}{a+b}$$

$$\mathrm{var}[\mu] = \frac{ab}{(a+b)^2 (a+b+1)}$$

Dirichlet distribution

$$f\left(x_1, \ldots, x_K; \alpha_1, \ldots, \alpha_K\right) = \frac{1}{\mathrm{B}(\boldsymbol{\alpha})} \prod_{i=1}^{K} x_i^{\alpha_i - 1}$$

$$\mathrm{B}(\boldsymbol{\alpha}) = \frac{\prod_{i=1}^{K} \Gamma(\alpha_i)}{\Gamma\left(\sum_{i=1}^{K} \alpha_i\right)}, \qquad \boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$$

$$\sum_{i=1}^{K} x_i = 1 \text{ and } x_i \geq 0 \text{ for all } i \in \{1, \ldots, K\}$$

Mahalanobis distance

$$\Delta^2 = (\mathbf{x} - \boldsymbol{\mu})^{\mathrm{T}} \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

Schur complement

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{M} & -\mathbf{M}\mathbf{B}\mathbf{D}^{-1} \\ -\mathbf{D}^{-1}\mathbf{C}\mathbf{M} & \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{M}\mathbf{B}\mathbf{D}^{-1} \end{pmatrix}, \quad \mathbf{M} = \left(\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}\right)^{-1}$$

Partitioned Gaussians

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}$$

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ba} & \boldsymbol{\Sigma}_{bb} \end{pmatrix}, \quad \boldsymbol{\Lambda} = \begin{pmatrix} \boldsymbol{\Lambda}_{aa} & \boldsymbol{\Lambda}_{ab} \\ \boldsymbol{\Lambda}_{ba} & \boldsymbol{\Lambda}_{bb} \end{pmatrix}$$

$$p\left(\mathbf{x}_a \mid \mathbf{x}_b\right) = \mathcal{N}\left(\mathbf{x} \mid \boldsymbol{\mu}_{a|b}, \boldsymbol{\Lambda}_{aa}^{-1}\right)$$

$$\boldsymbol{\mu}_{a|b} = \boldsymbol{\mu}_a - \boldsymbol{\Lambda}_{aa}^{-1}\boldsymbol{\Lambda}_{ab}\left(\mathbf{x}_b - \boldsymbol{\mu}_b\right)$$

$$p\left(\mathbf{x}_a\right) = \mathcal{N}\left(\mathbf{x}_a \mid \boldsymbol{\mu}_a, \boldsymbol{\Sigma}_{aa}\right)$$

population mean and population variance

$$\boldsymbol{\mu}_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n$$

$$\boldsymbol{\Sigma}_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}}\right)\left(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}}\right)^{\mathrm{T}}$$

$$\mathbb{E}\left[\boldsymbol{\mu}_{\mathrm{ML}}\right] = \boldsymbol{\mu}$$

$$\mathbb{E}\left[\boldsymbol{\Sigma}_{\mathrm{ML}}\right] = \frac{N-1}{N}\boldsymbol{\Sigma}$$

unbiased estimator of the covariance matrix

$$\widetilde{\boldsymbol{\Sigma}} = \frac{1}{N-1}\sum_{n=1}^{N}\left(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}}\right)\left(\mathbf{x}_n - \boldsymbol{\mu}_{\mathrm{ML}}\right)^{\mathrm{T}}$$

Bayesian inference for the Gaussian of μ

$$p(\mathbf{X} \mid \mu) = \prod_{n=1}^{N} p\left(x_n \mid \mu\right) = \frac{1}{\left(2\pi\sigma^2\right)^{N/2}}\exp\left\{-\frac{1}{2\sigma^2}\sum_{n=1}^{N}\left(x_n - \mu\right)^2\right\}$$

$$\mathbf{X} = \{x_1, \ldots, x_N\}$$

prior distribution of μ

$$p(\mu) = \mathcal{N}\left(\mu \mid \mu_0, \sigma_0^2\right)$$

posterior distribution of μ, where σ^2 is known

$$p(\mu \mid \mathbf{X}) \propto p(\mathbf{X} \mid \mu)p(\mu)$$

$$p(\mu \mid \mathbf{X}) = \mathcal{N}\left(\mu \mid \mu_N, \sigma_N^2\right)$$

$$\mu_N = \frac{\sigma^2}{N\sigma_0^2 + \sigma^2}\mu_0 + \frac{N\sigma_0^2}{N\sigma_0^2 + \sigma^2}\mu_{\mathrm{ML}}$$

$$\frac{1}{\sigma_N^2} = \frac{1}{\sigma_0^2} + \frac{N}{\sigma^2}$$

$$\mu_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}x_n$$

if $\sigma_0^2 \to \infty$, then

$$\mu_N = \mu_{\mathrm{ML}}$$
$$\sigma_N^2 = \sigma^2/N$$

Bayesian inference for the Gaussian of λ

$$p(\mathbf{X} \mid \lambda) = \prod_{n=1}^{N} \mathcal{N}\left(x_n \mid \mu, \lambda^{-1}\right) \propto \lambda^{N/2} \exp\left\{-\frac{\lambda}{2}\sum_{n=1}^{N}(x_n-\mu)^2\right\}$$

prior distribution of λ (gamma distribution)

$$\mathrm{Gam}(\lambda \mid a_0, b_0) = \frac{1}{\Gamma(a_0)} b_0^{a_0} \lambda^{a_0-1} \exp(-b_0\lambda)$$

$$p(\lambda \mid \mathbf{X}) \propto \lambda^{a_0-1}\lambda^{N/2} \exp\left\{-b_0\lambda - \frac{\lambda}{2}\sum_{n=1}^{N}(x_n-\mu)^2\right\}$$

posterior distribution of λ, where μ is known

$$p(\lambda \mid \mathbf{X}) = \mathrm{Gam}\left(\lambda \mid a_N, b_N\right)$$
$$a_N = a_0 + \frac{N}{2}$$
$$b_N = b_0 + \frac{1}{2}\sum_{n=1}^{N}(x_n-\mu)^2 = b_0 + \frac{N}{2}\sigma_{\mathrm{ML}}^2$$
$$\sigma_{\mathrm{ML}}^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n-\mu)^2$$

Bayesian inference for the Gaussian of μ and λ

$$p(\mathbf{X} \mid \mu, \lambda) = \prod_{n=1}^{N}\left(\frac{\lambda}{2\pi}\right)^{1/2} \exp\left\{-\frac{\lambda}{2}(x_n-\mu)^2\right\}$$
$$\propto \left[\lambda^{1/2}\exp\left(-\frac{\lambda\mu^2}{2}\right)\right]^N \exp\left\{\lambda\mu\sum_{n=1}^{N}x_n - \frac{\lambda}{2}\sum_{n=1}^{N}x_n^2\right\}$$

prior distribution of μ and λ

$$p(\mu, \lambda) \propto \left[\lambda^{1/2}\exp\left(-\frac{\lambda\mu^2}{2}\right)\right]^{\beta} \exp\{c\lambda\mu - d\lambda\}$$
$$= \exp\left\{-\frac{\beta\lambda}{2}(\mu - c/\beta)^2\right\}\lambda^{\beta/2}\exp\left\{-\left(d - \frac{c^2}{2\beta}\right)\lambda\right\}$$
$$p(\mu, \lambda) = \mathcal{N}\left(\mu \mid \mu_0, (\beta\lambda)^{-1}\right)\mathrm{Gam}(\lambda \mid a, b)$$
$$\mu_0 = c/\beta, \quad a = 1 + \beta/2, \quad b = d - \frac{c^2}{2\beta}$$

Bayesian inference for the multivariate Gaussian distribution for $\boldsymbol{\mu}$ known precision $\boldsymbol{\Lambda}$

$$\mathcal{N}\left(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1}\right)$$

Wishart prior distribution

$$\mathcal{W}(\boldsymbol{\Lambda} \mid \mathbf{W}, \nu) = B|\boldsymbol{\Lambda}|^{(\nu-D-1)/2} \exp\left(-\frac{1}{2} \operatorname{Tr}\left(\mathbf{W}^{-1}\boldsymbol{\Lambda}\right)\right)$$

$$B(\mathbf{W}, \nu) = |\mathbf{W}|^{-\nu/2} \left(2^{\nu D/2} \pi^{D(D-1)/4} \prod_{i=1}^{D} \Gamma\left(\frac{\nu+1-i}{2}\right)\right)^{-1}$$

normal-Wishart (Gaussian-Wishart) prior distribution

$$p\left(\boldsymbol{\mu}, \boldsymbol{\Lambda} \mid \boldsymbol{\mu}_0, \beta, \mathbf{W}, \nu\right) = \mathcal{N}\left(\boldsymbol{\mu} \mid \boldsymbol{\mu}_0, (\beta\boldsymbol{\Lambda})^{-1}\right) \mathcal{W}(\boldsymbol{\Lambda} \mid \mathbf{W}, \nu)$$

3 LINEAR MODELS FOR REGRESSION

simple linear regression

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1 x_1 + \ldots + w_D x_D$$

regression with basis functions $\boldsymbol{\phi}$

$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{j=1}^{M-1} w_j \phi_j(\mathbf{x})$$

$$y(\mathbf{x}, \mathbf{w}) = \sum_{j=0}^{M-1} w_j \phi_j(\mathbf{x}) = \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x})$$

dummy basis function

$$\phi_0(\mathbf{x}) = 1$$

Gaussian basis function

$$\phi_j(x) = \exp\left\{-\frac{(x-\mu_j)^2}{2s^2}\right\}$$

sigmoidal basis function

$$\phi_j(x) = \sigma\left(\frac{x-\mu_j}{s}\right)$$

$$\sigma(a) = \frac{1}{1+\exp(-a)}$$

$$\tanh(a) = 2\sigma(a) - 1$$

Gaussian noise model

$$t = y(\mathbf{x}, \mathbf{w}) + \epsilon$$

$$p(t \mid \mathbf{x}, \mathbf{w}, \beta) = \mathcal{N}\left(t \mid y(\mathbf{x}, \mathbf{w}), \beta^{-1}\right)$$

$$p(\mathbf{t} \mid \mathbf{X}, \mathbf{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}\left(t_n \mid \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}\left(\mathbf{x}_n\right), \beta^{-1}\right)$$

$$\begin{aligned} \ln p(\mathbf{t} \mid \mathbf{w}, \beta) &= \sum_{n=1}^{N} \ln \mathcal{N}\left(t_n \mid \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}\left(\mathbf{x}_n\right), \beta^{-1}\right) \\ &= \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) - \beta E_D(\mathbf{w}) \end{aligned}$$

where $E_D(\mathbf{w})$ is the sum-of-squares error function

$$E_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \left\{t_n - \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}\left(\mathbf{x}_n\right)\right\}^2$$

the maximum likelihood solution; differentiate with respect to $\mathbf{w}$

$$\nabla \ln p(\mathbf{t} \mid \mathbf{w}, \beta) = \sum_{n=1}^{N} \left\{ t_n - \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_n) \right\} \boldsymbol{\phi}(\mathbf{x}_n)^{\mathrm{T}}$$

$$0 = \sum_{n=1}^{N} t_n \boldsymbol{\phi}(\mathbf{x}_n)^{\mathrm{T}} - \mathbf{w}^{\mathrm{T}} \left(\sum_{n=1}^{N} \boldsymbol{\phi}(\mathbf{x}_n) \boldsymbol{\phi}(\mathbf{x}_n)^{\mathrm{T}} \right)$$

$$\mathbf{w}_{\mathrm{ML}} = \left(\boldsymbol{\Phi}^{\mathrm{T}} \boldsymbol{\Phi} \right)^{-1} \boldsymbol{\Phi}^{\mathrm{T}} \mathbf{t}$$

differentiate with respect to β

$$\frac{1}{\beta_{\mathrm{ML}}} = \frac{1}{N} \sum_{n=1}^{N} \left\{ t_n - \mathbf{w}_{\mathrm{ML}}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_n) \right\}^2$$

where $\boldsymbol{\Phi}$ is the design matrix; $\Phi_{nj} = \phi_j(\mathbf{x}_n)$ (matrix $_{\text{row, col}}$)

$$\boldsymbol{\Phi} = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \cdots & \phi_{M-1}(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & \cdots & \phi_{M-1}(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \cdots & \phi_{M-1}(\mathbf{x}_N) \end{pmatrix}$$

Moore-Penrose pseudo-inverse of the matrix $\boldsymbol{\Phi}$

$$\boldsymbol{\Phi}^{\dagger} \equiv \left(\boldsymbol{\Phi}^{\mathrm{T}} \boldsymbol{\Phi} \right)^{-1} \boldsymbol{\Phi}^{\mathrm{T}}$$

if $\boldsymbol{\Phi}$ is square and invertible, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

$$(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi})^{-1} = \boldsymbol{\Phi}^{-1}(\boldsymbol{\Phi}^{\mathrm{T}})^{-1}$$

$$(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^{\mathrm{T}} = \boldsymbol{\Phi}^{-1}((\boldsymbol{\Phi}^{\mathrm{T}})^{-1}\boldsymbol{\Phi}^{\mathrm{T}}) = \boldsymbol{\Phi}^{-1}\mathbf{I} = \boldsymbol{\Phi}^{-1}$$

$$\boldsymbol{\Phi}^{\dagger} \equiv \boldsymbol{\Phi}^{-1}$$

4 LINEAR MODELS FOR CLASSIFICATION

posterior distribution of discrete classes

$$p\left(\mathcal{C}_k \mid \mathbf{x}\right)=\frac{p\left(\mathbf{x} \mid \mathcal{C}_k\right) p\left(\mathcal{C}_k\right)}{p(\mathbf{x})}$$

transform the linear function of $\mathbf{w}$ using a nonlinear function $f(\cdot)$

$$y(\mathbf{x})=f\left(\mathbf{w}^{\mathrm{T}} \mathbf{x}+w_0\right)$$

two classes $\mathcal{C}_1$ and $\mathcal{C}_2$

$$y(\mathbf{x})=\mathbf{w}^{\mathrm{T}} \mathbf{x}+w_0$$

input vector $\mathbf{x}$ is assigned to class $\mathcal{C}_1$ if $y(\mathbf{x}) \geqslant 0$ and to class $\mathcal{C}_2$ otherwise.

In Euclidean space, two vectors are orthogonal iff their dot product is zero.

Consider two points $\mathbf{x}_{\mathrm{A}}$ and $\mathbf{x}_{\mathrm{B}}$ both of which lie on the decision surface, then $y\left(\mathbf{x}_{\mathrm{A}}\right)=y\left(\mathbf{x}_{\mathrm{B}}\right)=0$, $\mathbf{w}^{\mathrm{T}}\left(\mathbf{x}_{\mathrm{A}}-\mathbf{x}_{\mathrm{B}}\right)=0$ and hence the vector $\mathbf{w}$ is orthogonal to the decision surface.

the signed orthogonal distance of a general point $\mathbf{x}$ from the decision surface is given by $y(\mathbf{x}) /\|\mathbf{w}\|$.

proof of the above result
$\frac{\mathbf{w}}{\|\mathbf{w}\|}$ is the unit vector of $\mathbf{w}$
$\mathbf{x}_{\perp}$ is $\mathbf{x}$'s orthogonal projection onto the decision surface

$$\mathbf{x}=\mathbf{x}_{\perp}+r \frac{\mathbf{w}}{\|\mathbf{w}\|}$$

Multiplying both sides of this result by $\mathbf{w}^{\mathrm{T}}$ and adding w_0, and making use of $y(\mathbf{x})=\mathbf{w}^{\mathrm{T}} \mathbf{x}+w_0$ and $y\left(\mathbf{x}_{\perp}\right)=\mathbf{w}^{\mathrm{T}} \mathbf{x}_{\perp}+w_0=0$, $\mathbf{w}^{\mathrm{T}} \mathbf{w}=\|\mathbf{w}\|^2$ we have

$$r=\frac{y(\mathbf{x})}{\|\mathbf{w}\|}$$

compact representation $y(\mathbf{x})=\widetilde{\mathbf{w}}^{\mathrm{T}} \widetilde{\mathbf{x}}$ where $\widetilde{\mathbf{w}}=\left(w_0, \mathbf{w}\right)$ and $\widetilde{\mathbf{x}}=\left(x_0, \mathbf{x}\right)$.

a single K-class discriminant comprising K linear functions of the form

$$y_k(\mathbf{x})=\mathbf{w}_k^{\mathrm{T}} \mathbf{x}+w_{k 0}$$

assigning a point $\mathbf{x}$ to class $\mathcal{C}_k$ if $y_k(\mathbf{x})>y_j(\mathbf{x})$ for all $j \neq k$.

5 NEURAL NETWORKS

The linear models for regression and classification discussed in Chapters 3 and 4, respectively, are based on linear combinations of fixed nonlinear basis functions $\phi_j(\mathbf{x})$ and take the form

$$y(\mathbf{x}, \mathbf{w}) = f\left(\sum_{j=1}^{M} w_j \phi_j(\mathbf{x})\right)$$

This leads to the basic neural network model, which can be described a series of functional transformations. First we construct M linear combinations of the input variables $x_1, \ldots, x_D$ in the form

$$a_j = \sum_{i=1}^{D} w_{ji}^{(1)} x_i + w_{j0}^{(1)}$$

the superscript (1) indicates that the corresponding parameters are in the first "layer" of the network. We shall refer to the parameters $w_{ji}^{(1)}$ as weights and the parameters $w_{j0}^{(1)}$ as biases, following the nomenclature of Chapter 3.

The quantities a_j are known as activations. Each of them is then transformed using a differentiable, nonlinear activation function $h(\cdot)$ to give

$$z_j = h\left(a_j\right).$$

A sigmoid function is a bounded, differentiable, real function that is defined for all real input values and has a **non-negative derivative** at each point and **exactly one inflection point**. A sigmoid "function" and a sigmoid "curve" refer to the same object.

6 KERNEL METHODS

kernel function

$$k(\mathbf{x}, \mathbf{x}') = \boldsymbol{\phi}(\mathbf{x})^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}')$$

the kernel is a symmetric function of its arguments

$$k(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}', \mathbf{x})$$

reformulation of linear regression model with a regularized sum-of-squares error
dual representations

$$J(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^{N} \left\{ \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_n) - t_n \right\}^2 + \frac{\lambda}{2} \mathbf{w}^{\mathrm{T}} \mathbf{w}$$

solve for $\mathbf{w}$

$$\mathbf{w} = -\frac{1}{\lambda} \sum_{n=1}^{N} \left\{ \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_n) - t_n \right\} \boldsymbol{\phi}(\mathbf{x}_n) = \sum_{n=1}^{N} a_n \boldsymbol{\phi}(\mathbf{x}_n) = \boldsymbol{\Phi}^{\mathrm{T}} \mathbf{a}$$

$$a_n = -\frac{1}{\lambda} \left\{ \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_n) - t_n \right\}$$

where Φ is the design matrix, whose n^{th} row is given by $\boldsymbol{\phi}(\mathbf{x}_n)^{\mathrm{T}}$
substitute "$\mathbf{w} = \boldsymbol{\Phi}^{\mathrm{T}} \mathbf{a}$" into $J(\mathbf{w})$

$$J(\mathbf{a}) = \frac{1}{2} \mathbf{a}^{\mathrm{T}} \boldsymbol{\Phi} \boldsymbol{\Phi}^{\mathrm{T}} \boldsymbol{\Phi} \boldsymbol{\Phi}^{\mathrm{T}} \mathbf{a} - \mathbf{a}^{\mathrm{T}} \boldsymbol{\Phi} \boldsymbol{\Phi}^{\mathrm{T}} \mathbf{t} + \frac{1}{2} \mathbf{t}^{\mathrm{T}} \mathbf{t} + \frac{\lambda}{2} \mathbf{a}^{\mathrm{T}} \boldsymbol{\Phi} \boldsymbol{\Phi}^{\mathrm{T}} \mathbf{a}$$

Gram matrix ($\boldsymbol{\Phi} \in \mathbb{R}^{N \times M}$, $\mathbf{K} \in \mathbb{R}^{N \times N}$)

$$\mathbf{K} = \boldsymbol{\Phi} \boldsymbol{\Phi}^{\mathrm{T}}$$

$$K_{nm} = \boldsymbol{\phi}(\mathbf{x}_n)^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}_m) = k(\mathbf{x}_n, \mathbf{x}_m)$$

substitute "$\mathbf{K} = \boldsymbol{\Phi} \boldsymbol{\Phi}^{\mathrm{T}}$" into $J(\mathbf{a})$

$$J(\mathbf{a}) = \frac{1}{2} \mathbf{a}^{\mathrm{T}} \mathbf{K} \mathbf{K} \mathbf{a} - \mathbf{a}^{\mathrm{T}} \mathbf{K} \mathbf{t} + \frac{1}{2} \mathbf{t}^{\mathrm{T}} \mathbf{t} + \frac{\lambda}{2} \mathbf{a}^{\mathrm{T}} \mathbf{K} \mathbf{a}$$

solve for $\mathbf{a}$

$$\mathbf{a} = (\mathbf{K} + \lambda \mathbf{I}_N)^{-1} \mathbf{t}$$

substitute the solution into the linear regression model
prediction for a new input $\mathbf{x}$

$$y(\mathbf{x}) = \mathbf{w}^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x}) = \mathbf{a}^{\mathrm{T}} \boldsymbol{\Phi} \boldsymbol{\phi}(\mathbf{x}) = \mathbf{k}(\mathbf{x})^{\mathrm{T}} (\mathbf{K} + \lambda \mathbf{I}_N)^{-1} \mathbf{t}$$

$$\mathbf{k}(\mathbf{x}) \in \mathbb{R}^N, \text{ where } k_n(\mathbf{x}) = k(\mathbf{x}_n, \mathbf{x}) = \boldsymbol{\phi}(\mathbf{x}_n)^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{x})$$

kernel construction

$$k(\mathbf{x}, \mathbf{z}) = \left(\mathbf{x}^{\mathrm{T}}\mathbf{z}\right)^2$$

where

$$\begin{aligned} k(\mathbf{x}, \mathbf{z}) &= \left(\mathbf{x}^{\mathrm{T}}\mathbf{z}\right)^2 = (x_1 z_1 + x_2 z_2)^2 \\ &= x_1^2 z_1^2 + 2 x_1 z_1 x_2 z_2 + x_2^2 z_2^2 \\ &= \left(x_1^2, \sqrt{2} x_1 x_2, x_2^2\right) \left(z_1^2, \sqrt{2} z_1 z_2, z_2^2\right)^{\mathrm{T}} \\ &= \phi(\mathbf{x})^{\mathrm{T}} \boldsymbol{\phi}(\mathbf{z}) \end{aligned}$$

$$\phi(\mathbf{x}) = \left(x_1^2, \sqrt{2} x_1 x_2, x_2^2\right)^{\mathrm{T}}$$

compute $\left(\mathbf{x}^{\mathrm{T}}\mathbf{z}\right)^2$ instead of explicitly applying the transformations which is less computationally expensive

$$k\left(\mathbf{x}, \mathbf{x}'\right) \text{ is a valid kernel}$$

$$\leftrightarrow$$

the Gram matrix $\mathbf{K}$ is a positive semidefinite

for all possible choices of the set $\{\mathbf{x}_n\}$

7 SPARSE KERNEL MACHINES

8 GRAPHICAL MODELS

9 MIXTURE MODELS AND EM

objective function (distortion measure)

$$J = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \left\| \mathbf{x}_n - \boldsymbol{\mu}_k \right\|^2$$

where

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg\min_j \left\| \mathbf{x}_n - \boldsymbol{\mu}_j \right\|^2 \\ 0 & \text{otherwise} \end{cases}$$

differentiate the objective function J with respect to $\boldsymbol{\mu}_k$ for every k setting the derivative to zero

$$2 \sum_{n=1}^{N} r_{nk} \left(\mathbf{x}_n - \boldsymbol{\mu}_k \right) = 0$$

solve for $\boldsymbol{\mu}_k$

$$\boldsymbol{\mu}_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}}$$

objective function; where $\mathcal{V}\left(\mathbf{x}, \mathbf{x}'\right)$ is a general dissimilarity measure

$$\widetilde{J} = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \mathcal{V}\left(\mathbf{x}_n, \boldsymbol{\mu}_k\right)$$

Gaussian mixture model

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}\left(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)$$

10 APPROXIMATE INFERENCE

A central task in the application of probabilistic models is the evaluation of the posterior distribution $p(\mathbf{Z}|\mathbf{X})$ of the latent variables $\mathbf{Z}$ given the observed (visible) data variables $\mathbf{X}$, and the evaluation of expectations computed with respect to this distribution.

11 SAMPLING METHODS

the expectation of function $f(\mathbf{z})$ with respect to a probability distribution $p(\mathbf{z})$

$$\mathbb{E}[f] = \int f(\mathbf{z})p(\mathbf{z})\mathrm{d}\mathbf{z}$$

expectation approximation by a finite sum

$$\widehat{f} = \frac{1}{L}\sum_{l=1}^{L} f\left(\mathbf{z}^{(l)}\right)$$

12 CONTINUOUS LATENT VARIABLES